

ESP32 Smart Energy Monitoring and Automatic Load Management System

Schematic Diagram

Project Overview

This project implements a real-time energy monitoring and automatic load management system using an ESP32 microcontroller. Voltage and current levels are simulated using potentiometers and displayed on an OLED screen, while a relay, buzzer, and status LEDs provide automatic load control and visual/audible alerts based on the detected consumption level.

Wiring Connections Table

Component	Pin	ESP32 GPIO Connection
ESP32 DevKit V1	—	Main microcontroller board
OLED SSD1306 (I2C, 0x3C)	VCC	3V3
OLED SSD1306 (I2C, 0x3C)	GND	GND
OLED SSD1306 (I2C, 0x3C)	SCL	GPIO 22
OLED SSD1306 (I2C, 0x3C)	SDA	GPIO 21
Potentiometer 1 (Voltage)	VCC	3V3
Potentiometer 1 (Voltage)	GND	GND
Potentiometer 1 (Voltage)	SIG	GPIO 34 (ADC)
Potentiometer 2 (Current)	VCC	3V3
Potentiometer 2 (Current)	GND	GND
Potentiometer 2 (Current)	SIG	GPIO 35 (ADC)
Relay Module	VCC	5V
Relay Module	GND	GND
Relay Module	IN	GPIO 19
Piezo Buzzer	Pin 1 (–)	GND
Piezo Buzzer	Pin 2 (+)	GPIO 23
Green LED	Anode (A)	GPIO 25 (via R1, 220 Ω)
Green LED	Cathode (C)	GND (through R1)
Yellow LED	Anode (A)	GPIO 26 (via R2, 220 Ω)
Yellow LED	Cathode (C)	GND (through R2)
Red LED	Anode (A)	GPIO 33 (via R3, 220 Ω)

Component	Pin	ESP32 GPIO Connection
Red LED	Cathode (C)	GND (through R3)
Resistor R1 (220 Ω)	1 / 2	Green LED Cathode → GND
Resistor R2 (220 Ω)	1 / 2	Yellow LED Cathode → GND
Resistor R3 (220 Ω)	1 / 2	Red LED Cathode → GND
Power Supply	3V3	OLED, Potentiometer 1, Potentiometer 2
Power Supply	5V	Relay Module
Ground (Common)	GND	OLED, Potentiometers, Relay, Buzzer, LEDs (via resistors)

ESP32 GPIO Pin Configuration Table

ESP32 GPIO	Connected To	Function
GPIO 21	OLED SSD1306 (SDA)	I2C Data Line
GPIO 22	OLED SSD1306 (SCL)	I2C Clock Line
GPIO 34	Potentiometer 1 (SIG)	Voltage Sensing Input (ADC, input-only)
GPIO 35	Potentiometer 2 (SIG)	Current Sensing Input (ADC, input-only)
GPIO 19	Relay Module (IN)	Load ON/OFF Control Output
GPIO 23	Piezo Buzzer	Overload Audible Alert Output
GPIO 25	Green LED (via R1)	Normal Mode Indicator
GPIO 26	Yellow LED (via R2)	High Consumption Mode Indicator
GPIO 33	Red LED (via R3)	Overload Mode Indicator
3V3	OLED, Potentiometer 1, Potentiometer 2	3.3 V Power Supply
5V	Relay Module	5 V Power Supply
GND	All Modules (Common Ground)	Ground Reference

Operating Modes

Normal Mode

Indicator	State
Green LED	ON
Relay	ON
Buzzer	OFF
OLED Display	Shows "NORMAL"

High Consumption Mode

Indicator	State
Yellow LED	ON
Relay	ON
Buzzer	OFF
OLED Display	Shows "HIGH"

Overload Mode

Indicator	State
Red LED	ON
Relay	OFF
Buzzer	ON
OLED Display	Shows "OVERLOAD"

Notes

This project is developed using the ESP32 DevKit V1 microcontroller and simulated in Wokwi. Voltage and current inputs are represented using potentiometers for simulation purposes, and GPIO pin assignments in this document are derived directly from the project's Wokwi **diagram.json** configuration file.